UNDERSTANDING GAS LAWS

Name_________________________________________ period__________ desk #___________

In the boxes below, draw a diagram of what you think ten gas particles would look like if you could zoom really close in to see them. Use ● for particles, → to show their movement. Bigger arrows mean more velocity. The box is the container.

Low temperature    High temperature    High volume    low volume

High pressure    Low pressure

**Kinetic Molecular Theory:**

Go to [http://phet.colorado.edu/en/simulation/gas-properties](http://phet.colorado.edu/en/simulation/gas-properties)

Select the Ideal Simulation

Click to hold the Volume constant.

1. Use the pump to put one pump of gas into the box.

a. What happens to the clump of particles?_____________________________________________

(To answer the following questions, keep your eye on one particle and notice how it moves.)

b. How do the particles move? (straight line, circular, random, etc.)______________________________

c. Do the particles stay at a constant speed?_____________

d. If not, what causes the speed to change?______________________________________________

f. Do they always move in the same direction?________________

g. If not, what causes their direction to change?____________________________________________

2. Using the Particles setting on the right side of the screen, put 100 “heavy species” in the container. Give it time for the pressure to stabilize. Observe the motion of the particles.

a. Record the pressure______________ ( The number will jump around- choose an average.)

b. Reset the number of “heavy species” to zero, and the “light species” to 100. Observe the motion of the particles.

c. Record the pressure________________

d. Does the mass of the particles significantly affect the pressure of the container? _______________

e. Explain this using your observations.______________________________________________________________

______________________________________________________________________________________________

**Partial Pressures**

3. Put 100 of “heavy species” and no “light species”.

a. Record the pressure. ______________

b. Put 50 of the “light species” and no “heavy species”

c. Record the pressure___________________

d. Put 50 “light species” AND 100 “heavy species” together. Record the pressure_________________

e. How does this compare to the pressures from 3a. and 3c?___________________________________

f. What can you conclude about the relationship between the partial and total pressure?

_______________________________________________________________________________________

**Boyle’s Law:**

Since Boyle’s Law deals with pressure and volume, temperature must be constant.

1. On the Hold Constant box in the top right, select temperature to be constant.

2. Place 200 “heavy species” in your container.

3. Use the handle on the left to change the volume of the container.

a. What happens to the pressure as the volume changes?__________________________________________

As the volume goes_______________ the pressure goes_____________________.

This is a(n) _______________________ relationship.

4. Diagram the particles in the boxes that would
Model Boyle’s Law. (Include arrows.)
Label the variables below each box.

5. Experiment by changing the number of species, volume and pressure. What combination do you need to blow the top off?______________________________________________________________________________________

**Charles' Law: (Note: For this simulation you have to put the species in the container BEFORE you set the constant parameter. .**

Since Charles' Law deals with temperature and volume, _________________ must be constant.

1. Place 200 "heavy species" in your container

2.. On the Hold Constant box on the right, select Pressure ↕V to be constant.

3. Use the heat/cold slider on the bucket at the bottom to heat up the container.

a. What happens to the volume as the temperature changes?__________________________

b. As the temperature goes______________ the volume goes____________________.

c. This is a(n) ______________________ relationship.

4. Diagram the particles in the boxes that would model Charles' Law. (Include arrows.)
Label the variables below each box.

5. Experiment by changing the temperature and volume. What combination do you need to blow the top off?___________________________________________________________________________________

**Gay-Lussac's Law:**

Since Gay-Lussac's Law deals with pressure and temperature, ____________________ must be constant.

1. On the Hold Constant box in the top right, select the appropriate constant.

2. Place 200 "heavy species" in your container.

3. Use the Heat/Cold toggle in the bucket on the bottom to change the temperature of the container.

a. What happens to the pressure as the temperature changes?__________________________

As the temperature goes______________ the pressure goes____________________.

This is a(n) ______________________ relationship.

4. Diagram the particles in the boxes that would model Gay-Lussac's Law. (Include arrows.)
Label the variables below each box.

5. Experiment by changing the number of species, temperature and pressure. What combination do you need to blow the top off?____________________________________________________________